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Formation of fluorine molecule (F_2):

Valence shell electronic configuration of fluorine atom : $2s^2\ 2p_x^2,\ 2p_y^2,\ 2p_z^1$

When the half filled p_z orbitals of two fluorine overlaps along the z-axis, a σ-covalent bond is formed between them.

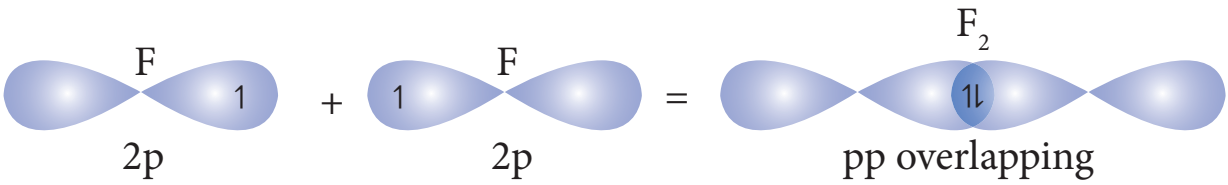


Fig 10. 19 Formation of F_2 Molecule

Formation of HF molecule:

Electronic configuration of hydrogen atom is $1s^1$

Valence shell electronic configuration of fluorine atom : $2s^2\ 2p_x^2,\ 2p_y^2,\ 2p_z^1$

When half filled 1s orbital of hydrogen linearly overlaps with a half filled $2p_z$ orbital of fluorine, a σ-covalent bond is formed between hydrogen and fluorine.

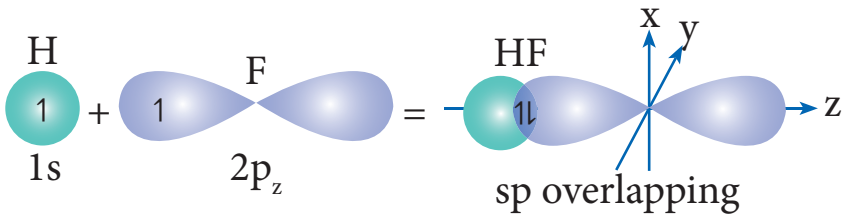
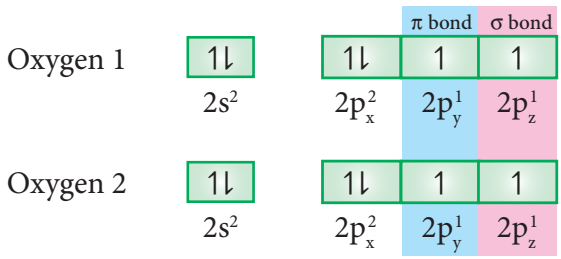


Fig 10.20 Formation of HF Molecule

Formation of oxygen molecule (O_2):

Valence shell electronic configuration of oxygen atom : $2s^2\ 2p_x^2,\ 2p_y^1,\ 2p_z^1$



When the half filled p_z orbitals of two oxygen overlaps along the z-axis (considering molecular axis as z axis), a σ-covalent bond is formed between them. Other two half filled p_y orbitals of two oxygen atoms overlap laterally (sideways) to form a π-covalent bond between